DRAFT: Comparing Habitat Suitability Modeling Approaches for the Eastern Oyster in the Pensacola Bay System

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Introduction

The eastern oyster (*Crassostrea virginica*) are viewed as a foundational species as they are directly responsible for creating and providing numerous ecosystem goods and services such as food and habitat production, shoreline stabilization and protection, water quality improvement, nutrient cycling, and carbon sequestration (Geselbracht et al. 2024; Grabowski et al. 2012; Barnes et al. 2007). Additionally, they provide numerous economic benefits such as supporting oyster fisheries, recreational harvesting, tourism, along with the associated employment that goes along with those industries (Geselbracht et al. 2024; Peyre et al. 2021). In one study, Grabowski et al. (2012) estimated the average annual value to range from $10,325 to $99,421 per hectare (2011 Dollars).

In recent decades, oyster reefs have experienced substantial declines, with only an estimated 15% remaining worldwide, and 50 to 80% being lost within the Gulf of (America) alone (Pollack et al. 2012; Peyre et al. 2021). The decline is largely attributed to a combination of anthropogenic activities such us over harvesting, habitat alteration, sedimentation and water quality decline, disease and changing hydrology (Peyre et al. 2021; Geselbracht et al. 2024). In response to these sharp declines, many restoration efforts have been undertaken with varying strategies and success rates; recently, planning documents released by the Deepwater Horizon Natural resource Damage Assessment Trustees called for large-scale development of reef networks to increase resilience (Geselbracht et al. 2024).

Oyster decline in the Pensacola Bay System

On such system to experience oyster decline is the Pensacola Bay System (PBS). The PBS is a large estuarine system located in Escambia and Santa Rosa counties in northwest Florida.

The system is comprised of multiple waterbody's including Pensacola Bay, Escambia Bay, Blackwater Bay, East Bay, and Santa Rosa Sound; and is influenced heavily by freshwater input from the Escambia, Blackwater and Yellow Rivers (Christensen et al. 1997). Historically, the PBS supported extensive oyster reefs that provided critical habitat for a variety of species and substantial economic value to the surrounding communities (Christensen et al. 1997). However, over the past several decades, the extent of oyster habitat within the PBS has declined significantly due to a combination of factors including over harvesting, habitat degradation, and water quality issues. With some estimates indicating a decline of 15,000ha to just 91.5 ha of oyster habitat between the years of 1883 and 2021 (Geselbracht et al. 2024). This decline has prompted efforts to restore oyster reefs within the PBS (Pensacola and Perdido Bays Estuary Program n.d.; The Nature Conservancy 2021), with multiple studies aiming to identify the most suitable areas within the estuary to focus restoration efforts (Geselbracht et al. 2024; Christensen et al. 1997; Pensacola and Perdido Bays Estuary Program n.d.).

With limited resources available for restoration, it is critical to identify areas where chances of oyster survival and long-term reef establishment are at their highest, and where poor site selection could ultimately lead to unsatisfactory results. As with any ecosystem, the environmental conditions within the PBS vary greatly in time and space, and such variations can have great impacts on oyster survival and establishment. Therefore, it is important to understand the environmental factors that drive oyster occurrence so that scientist and policy makers can make informed decisions regarding said restoration site selection (Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024).

Environmental controls on Eastern oyster habitat

Oyster recruitment, survival, and reef establishment are influenced by complex interactions between numerous physical, chemical, and biological environmental factors. As such, determining the most influential factors is a multivariate problem that requires careful consideration of each factor and its interactions with others (Peyre et al. 2021; Pollack et al. 2012; Barnes et al. 2007; Christensen et al. 1997). To help narrow the focus, the most commonly cited environmental factors were examined in the literature and are summarized here.

Perhaps the most influential and widely reported environmental driver of oyster suitability is salinity. Representing an important environmental gradient within estuarine systems that influences physiology, recruitment, disease, predation, and survival. Other important characteristics often reported can be seen in Table 1 and include: dissolved oxygen, temperature, depth, turbidity, and bottom type (substrate) (Peyre et al. 2021; Pollack et al. 2012; Barnes et al. 2007; Geselbracht et al. 2024; Christensen et al. 1997).

Table 1: Common environmental drivers of Eastern Oyster habitat suitability.

Factor	Description	Sources
Salinity	Controls feeding, growth, reproduction, recruitment, disease, predation, and survival.	(Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997;Geselbracht et al. 2024; Peyre et al. 2021; Pollack et al. 2012)
Temperature	Regulates metabolism, filtration, growth, spawning, and larval development.	(Barnes et al. 2007; Christensen et al. 1997; Geselbracht et al. 2024;Peyre et al. 2021; Pollack et al. 2012)
Substrate type	Stable hard substrate supports larval attachment and reef persistence, whereas soft or rapidly accumulating sediments can bury or destabilize oysters.	(Barnes et al. 2007; Birch et al. 2021; Christensen et al. 1997;Geselbracht et al. 2024; Peyre et al. 2021)
Dissolved oxygen	Adequate oxygen supports respiration, feeding, and growth.	(Christensen et al. 1997; Geselbracht et al. 2024; Peyre et al. 2021;Pollack et al. 2012)
Hydrology / water movement	Water movement regulates salinity and transports larvae, oxygen, and food.	(Barnes et al. 2007; Geselbracht et al. 2024; Peyre et al. 2021)
Depth / bathymetry	Depth influences exposure, temperature, oxygen conditions, and restoration feasibility.	(Christensen et al. 1997; Geselbracht et al. 2024; Pollack et al. 2012)
Turbidity / sedimentation	High suspended-sediment loads can interfere with filtration and settlement. Sediment deposition can smoother oyster reefs.	(Peyre et al. 2021; Pollack et al. 2012)
Food availability / chlorophyll a	Chlorophyll a is often used as a proxy for food availability.	(Geselbracht et al. 2024; Peyre et al. 2021)

It should be recognized that not all environmental factors are adequately spatially distributed or represented within the available datasets, and therefore careful consideration should be given when selecting the final dataset used for modeling habitat suitability (Peyre et al. 2021; Geselbracht et al. 2024; Christensen et al. 1997).

Species Distribution Modelling

Quantifying the relationships between the distribution of species and the species-environment interactions have a long history in ecological research, having been described as far back as the late 17th early 18th century in which researchers observed biological patterns in terms of their relationships with geographical and/or environmental gradients (Rushton et al. 2004; Elith and Leathwick 2009). Since then, the propagation of large-scale spatial datasets, the development and integration of GIS, increased access to statistical packages and computational power have allowed for much more development in modeling complex species-environment interactions (Johnson and Gillingham 2005). Out of such developments has arisen a technique known as Species Distribution Modeling (SDM), which can be loosely defined as numerical tools that combine observations of species occurrence or abundance with environmental estimates (Elith and Leathwick 2009). According to Elith and Leathwick (2009), SDMs can be largely classified within several families of models: expert-driven or heuristic, environmental-envelope, environmental-similarity, regression-based, Bayesian and hierarchical, machine-learning and data mining, and ensemble or multimodal. Additionally, the selection of an SDM approach largely depends on a multitude of factors such as: the study's objective, quality and quantity of available data, ecological characteristics of the species, spatial and temporal scale, whether input data includes absence data or presence-only data (Elith and Leathwick 2009). With so many choices of modeling technique and the driving factors behind that choice, deciding on one model can be quite difficult and the choice usually defaults to a standard technique depending on the subject matter.

Habitat Suitability Indices

One popular approach used in many studies involving the Eastern oyster throughout the gulf region is that of the expert-driven Habitat Suitability Index (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Christensen et al. 1997). HSIs are typically used in situations where quantitative data on species distributions is difficult to collect or non-existent, relying on expert-driven variable selection, meta-analysis, and classification methods that transform environmental characteristics into standardized estimates of habitat suitability (Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024; Johnson and Gillingham 2005).

Within the PBS, HSIs have been used to develop GIS-based and estuary-specific HSMs that integrate biological, physical, chemical, and management-related variables in order to identify suitable restoration areas with the system. In the mid 1990's Christensen et al. (1997) developed a HSI for the eastern oyster in the PBS to demonstrate the model frameworks usefulness in illustrating species suitability in a system wide approach. Geselbracht et al.

(2024) then followed and further expanded on the approach incorporating management factors into the model, adding a layer of regulatory influence to the model output versus the purely environmental analysis of the prior. Additionally, the Pensacola and Perdido Bay Estuary Program recently built on-top of Geselbracht et al. (2024) by incorporating some of their own data and refining some of the weighting (Pensacola and Perdido Bays Estuary Program n.d.).

These previous studies alone highlight the strengths and weakness of the HSI approaches. Some of the strengths include interoperability, adaptability, and the integration of ecological knowledge, but the results entirely rely on expert decisions regarding variable selection, suitability functions, weighting and scoring, and how to aggregate the results. All of these present opportunity for differing outputs even if the same inputs were used by different groups (Geselbracht et al. 2024; Johnson and Gillingham 2005).

Maximum Entropy Modelling (MaxEnt)

MaxEnt is a presence-background SDM framework based on general-purpose machine learning methods that are useful for making predictions using incomplete information by relating environmental conditions at known occurrence locations to the environmental conditions throughout the study area (Phillips et al. 2006; Elith et al. 2011). Some of the main benefits of MaxEnt is that it uses presence-only data, reducing the need for the often un-available absence data, it has the ability to model complex interactions between environmental conditions, and it has demonstrated high predictive performance while limiting over fitting through regularization (Phillips et al. 2009; Merow et al. 2013; Elith et al. 2011). An added benefit to this particular study is the fact that MaxEnt is tailored towards presence-only species occurrence data, making it highly suitable for the data set used here in which the FWC-FWRI (2026) datasets explicitly collected occurrence data without including absence locations.

However, the use of such algorithms should not be made blindly as many issues can arise if careful precautions are not taken. Due to the nature of ecological many biases in the data samples can arise such as spatial and temporal autocorrelation, group dependence structures, hierarchical structures and multicollinearity leading to overfitting model parameters and over-optimistic prediction confidence (Dormann et al. 2013; Kass et al. 2021; Roberts et al. 2017). Thankfully, significant work has been put into this through cross-validation techniques and the development of model evaluation software packages such as ENMeval 2.0 implemented through the R statistical language (Kass et al. 2021; Roberts et al. 2017)

Study Objective

To demonstrate how a MaxEnt SDM can be used as a complimentary tool to help researchers and practitioners leading ecological restoration activities, this study aims to develop an empirically fitted MaxEnt habitat-suitability model using observed oyster occurrence data and known environmental conditions. The primary objective will be to develop a spatially explicit SDM representing relative Eastern oyster habitat suitabiliy throughout the PBS. Next, the resulting suitability surface will be evaluated agains the existing Pensacola Bay HSI developed by

Geselbracht et al. (2024) and iterated upon by Pensacola and Perdido Bays Estuary Program (n.d.). To achieve this, environmental predictors that are commonly associated with oyster habitat suitability will be identified and obtained. The predictors will be used to fit the MaxEnt model and model performance will be evaluated using spatially separated cross-validation. Finally, we will compare the predicted surface to the existing HSI to determine areas of agreement and disagreement, and we will use these results to determine whether the MaxEnt framework provides an empirical alternative or compliment to HSI-based restoration planning. It is hypothesized that the MaxEnt model will successfully identify patterns of habitat suitability for the Easter oyster withinthe PBS. Overall agreement between the HSI and MaxEnt is expected to be high. However, MaxEnt is expected to produce a smoother, more continuous suitability surface because of the empirically fitted environmental relationships as opposed to predefined, lower-resolution HSI suitability envelopes. As a result, the MaxEnt model could prove beneficial to help narrow down project locations to help reinforce expert-driven model outputs.

Methods

All data analysis and processing was performed using the R programming language (R Core Team 2024). Environmental observation data was acquired from the EPA's Water Quality Portal via the dataRetrieval r-package (USGS 2026; DeCicco et al. 2026). Data QA/QC and preprocessing was completed utilizing the Tidyverse package and any spatially aware data processing was conducted using the package sf (Wickham et al. 2019; Pebesma 2018). All spatial data was projected to the NAD83 / Florida North State Plane Coordinate System (EPSG: 2238) which minimizes spatial distortion in the study areas geographic region.

Study Area

The study area for this project is the Pensacola Bay System which includes portions of Pensacola Bay, Escambia Bay, East Bay and Blackwater Bay. Boundaries of this study area were achieved by accessing Florida Department of Protections (FDEP) Waterbod IDs feature layer hosted on ArcGIS Online (FDEP 2005). Once retrieved, the dataset was filtered to include only relevant WBIDs (WBID = 548) and then combined to create a singular study area polygon. Creating the study area polygon is an important first step as it was needed to extract relevant environmental data from much larger datasets in the coming data gathering. The study area was intentionally chosen in an attempt to match the previous study area applied in Geselbracht et al. (2024).

Data Acquisition and Processing

Oyster occurence data was acquired from the Florida Fish and Wildlife Commission's GIS portal as polygons and converted to points using polygon centroids (FWC-FWRI 2026). Salinity, dissolved oxygen, temperature, and turbidity, and chlorophyll-a observations were obtained from the Water Quality Portal using the 'dataRetrieval' package in R for all sites located

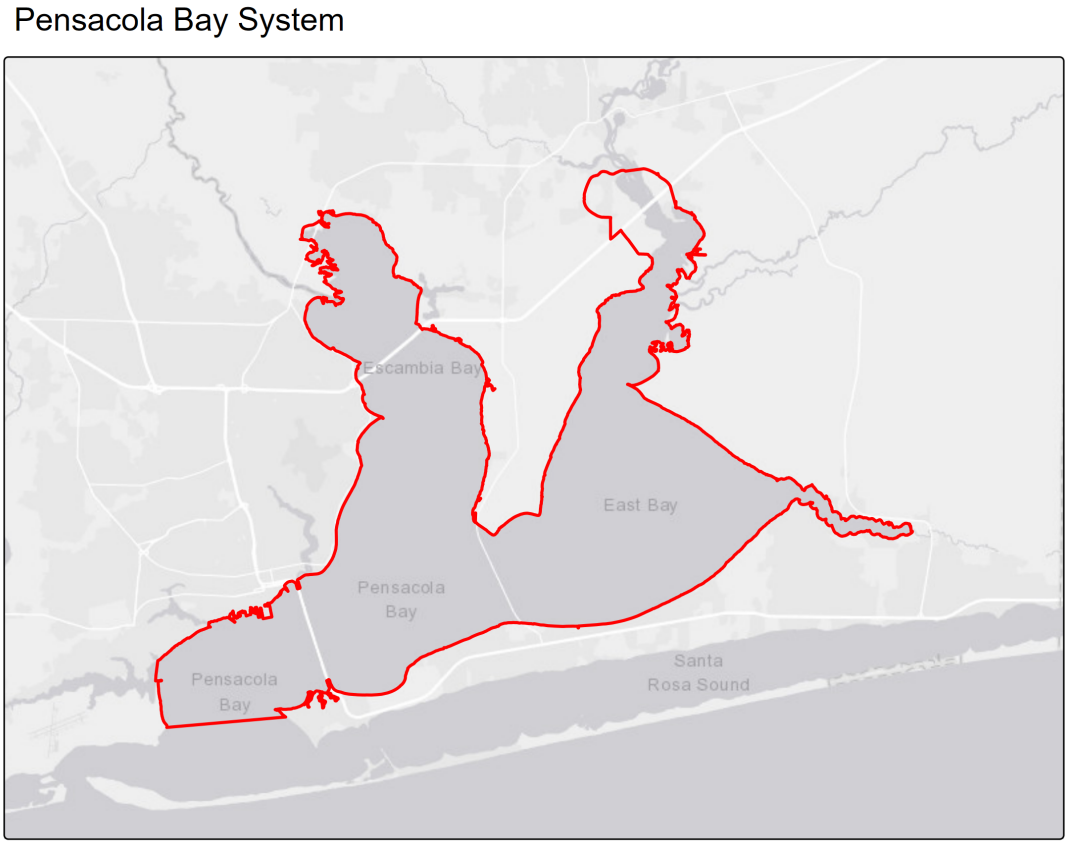


Figure 1: Study area for the Pensacola Bay System.

within the PBS and collected within the 20 year period from 2005 to 2025 DeCicco et al. (2026). Parameters were chosen based on relevance and represenation in previous oyster habitat retoration studies (Peyre et al. 2021; Barnes et al. 2007; Pollack et al. 2012; Geselbracht et al. 2024). Bathymetry data for the PBS was obtained from NOAA (National Centers for Environmental Information 2018). Bottom sediment data was obtained from the Pensacola and Perdido Bay Estuary Program.

Table 2: Environmental Parameters

Dataset	Data Type	# of Records	Processed Count	Source
Salinity	Point	6992	95	(USGS 2026)
Water Temperature	Point	7159	100	(USGS 2026)
Substrate type	Raster	NA	NA	(Pensacola and Perdido Bays Estuary Program n.d.)
Dissolved oxygen	Point	6261	99	(USGS 2026)
Depth / bathymetry	Raster	NA	NA	(USGS 2026)
Turbidity	Point	3402	75	(USGS 2026)
Chlorophyll a	Point	1847	44	(USGS 2026)
Oyster Reefs	Polygon	443	443	(FWC-FWRI 2026)

Water-quality Data Preprocessing

All water quality parameters were inspected to ensure comparable measurement units and analytical procedures and avoid combining different forms of similar analytes. Further, to maintain consistency with other similar oyster studies, all data was filtered to only include warm season conditions. Here, the warm season is defined as any data that was collected in the monthes of May, June, July, August, September, or October. Finally, the data was screened for any extreme outliers, where outliers were defined as:

$$\begin{aligned} \text{Outlier if} \quad & x < Q_{0.25} - 1.5(IQR) \\ \text{or} \quad & x > Q_{0.75} + 1.5(IQR) \end{aligned}$$

$$\text{where } IQR = Q_{0.75} - Q_{0.25}$$

To avoid spatial bias during the interpolation process, station-level site data was aggregated using 2km hexagonal spatial bins, taking the mean value of all data within that bin and returning a single value for each hexagon, using the hexagons centroid as the representative point for the data value. Stations that only had one sample were removed.

Environmental Prediction Rasters

Prior to use in the MaxEnt modeling framework, continuous prediction rasters were created for dissolved oxygen, salinity, turbidity, and water temperature using the Inverse Path Distance Weighting (IPDW) package in R (Stachelek 2023). IPDW is a variant of Inverse Distance Weighting (IDW), which is a deterministic interpolation method that allows for the estimation of point values using the equation:

$$V = \frac{\sum_{i=1}^{n} v_i d_i^{-p}}{\sum_{i=1}^{n} d_i^{-p}}$$

where d is the distance between the prediction and measurement points, v_i is the parameter value, and p is a power parameter. However, in IPDW d is a calculated path-distance value rather than a typical euclidean distance as used in IDW (Stachelek and Madden 2015).

IPDW was selected due to the model's ability to be barrier aware and avoid interference or influence from crossing land boundaries that can be introduced when using more traditional euclidean based interpolation methods (Stachelek and Madden 2015). This is especially important given the geography on PBS where the Garcon Point area is a narrow, low lying peninsula separating Escambia Bay to the west and Blackwater Bay to the east. Both of which are significantly influence by different riverine inputs. Instead, IPDW weighs observations according to the shortest water-path distance through the aquatic environment.

IPDW parameter sensitivity analysis

All parameter IPDW models were evaluated for sensitivity to determine influence of distance power and maximum path distance on overall interpolation performance. In IPDW, distance power determines how rapid the site influence decays as path distance increases. The distance powers (p) tested were: 0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 2.00, 2.50 and 3.00. The influence of maximum path distance was also evaluated for each IPDW model, in which the maximum range determines the max distance in which a measured value can contribute to the prediction value. Maximum ranges evaluated were: 5km, 10km, 15km, and 20km. All combinations of distance power and maximum range values were evaluated to determine the best fit for each parameter.

Spatial cross-validation was used to determine interpolation accuracy using the blockCV package in R, which uses spatial folds to split data into test and training data sets (Valavi et al. 2019). To improve fitting, the fold-generation procedure was repeated, each time refitting the IPDW model and comparing the results to the with-held observation spatial-blocks. For this analysis 5 spatial folds were used with 5km spatial blocks, and blocks were assigned the folds randomly. This process was then repeated across 5 random seeds in which a new spatial partition was generated for each seed. For each spatial fold, the monitoring locations were used in the training data to construct the IPDW prediction surface. Withheld monitoring locations

were used to extract predicted values from the prediction raster and compared to the observed station means.

Finally, model performance was evaluated using calculated prediction-error statistics for each distance power - max range combination. Each statistic was summarized across each repetition to generate a mean Mean Absolute Error (MAE), the standard deviation of the MAE, the mean Root Mean Square Error (RMSE) and its standard deviation, the mean prediction bias (+/-) and finally the prediction coverage. Mean MAE was the primary metric used in parameter selection, although in cases were differences between several of the lowest results were negligible, the smoother surfaces were given preference to better reflect the reality of a well mixed natural system.

The final prediction surfaces were generated by fitting the IPDW algorithm using the selected distance power and max range values on the full dataset for each environmental variable. A new path-distance raster was also created using the selected maximum range value from the sensitivity analysis. The final prediction surfaces were validated using a reproducible known seed, 5km spatial blocks, and five folds to calculate the final MAE, RMSE, prediction bias, and prediction coverage.

Max-Ent Species Distribution Modeling

Species distribution modeling was performed using the MaxEnt framework within the 'dismo' package in R (Hijmans et al. 2024).

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Results

Water Quality Observations

Queried water quality observations for a twenty year period between 2005 and 2025 returned 25,665 observations from 174 unique sampling sites located within the PBS (Table 3). Water temperature returned the largest number of records (7,161), followed by salinity (6,994), dissolved oxygen (6,261), turbidity (3,402), and chlorophyll-a (1,847). Chlorophyll-a only had measurements at 54 stations compared to the 146-169 for the other parameters.

Table 3: Summary of raw water-quality data available for each environmental parameter.

Parameter	Total Observations	Unique Stations	Reported Units
Water temperature	7,161	169	deg C
Salinity	6,994	159	ppt, ppth, PSS
Dissolved oxygen	6,261	169	mg/L

Table 3: Summary of raw water-quality data available for each environmental parameter.

Parameter	Total Observations	Unique Stations	Reported Units
Turbidity	3,402	146	NTU
Chlorophyll-a	1,847	54	ug/L

After processing and spatial aggregation, there were a total of 40 records each representing the mean value for that location (Table 4). Chlorophyll-a processing resulted in the fewest 754 observations summarized within 27 spatial bins using 48 stations. Dissolved oxygen had 3072 observations summarized into 68 bins across 145 stations. Salinity had 3,400 observations summarized into 67 bins with 135 stations, and turbidity retained 1563 observations within 61 spatial bins using 122 unique stations. Chlorophyll-a was excluded from the final prediction rasters due to poor spatial distribution and limited data availability.

Table 4: Summary of processed warm-season water-quality data used for spatial interpolation.

Parameter	Spatial bins	Unique Stations	Total Observations	First year	Last year	Units
Chlorophyll-a	27	48	754	2005	2024	µg/L
Dissolved oxygen	68	145	3,072	2005	2024	mg/L
Salinity	67	135	3,400	2005	2024	ppt
Turbidity	61	122	1,563	2005	2024	NTU
Water temperature	68	145	3,441	2005	2024	°C

Water Quality Mean Observation Data Distribution

Spatially binned mean Chlorophyll-a values ranged from 3.27 to 63.8ug/L with an average mean of 11.24ug/L. The distribution was heavily skewed to the right due to an extreme outlier of 60ug/L. Dissolved Oxygen was normally distributed with values ranging from 5.11 to 7.99mg/L with an average mean of 11.24ug/L. Salinity was normally distributed with a minimum value of 2.31 to 26.73 with an average mean of 16.18PPT. Turbidity was right skewed and had a minimum value of 0.03 to 9.62 with an average mean of 3.15NTU. Water temperature was highly variable and had a minimum value of 25.93 to 30.43 with an average mean of 27.84degrees C.

IPDW Interpolation Sensitivity and Prediction

Model sensitivity was evaluated for each parameter evaluating 36 combinations of max range and distance power values using five repititions of five-fold spatial cross-validation Figure 4.

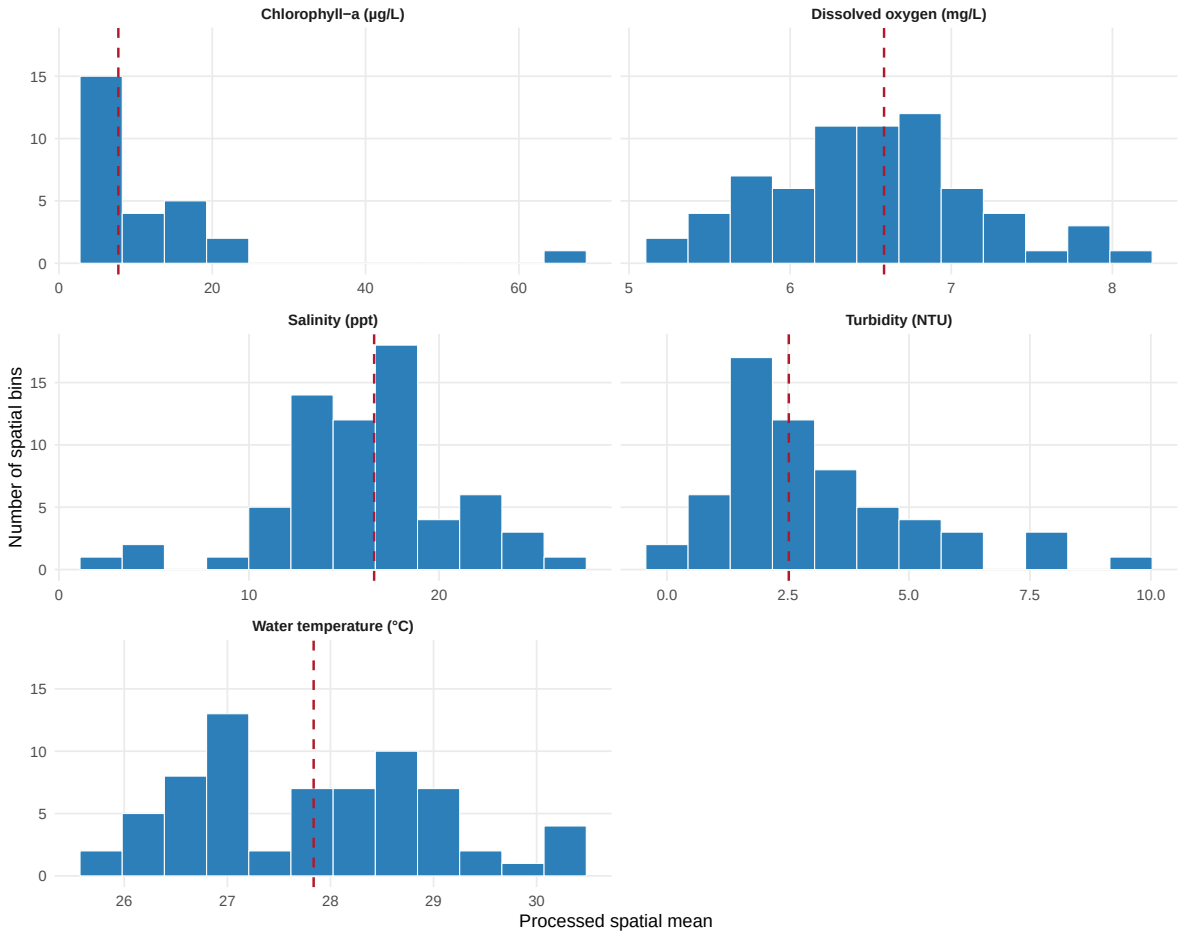


Figure 2: Distribution of processed warm-season spatial-bin means for each environmental parameter. Dashed lines identify the median of each distribution.

Following evaluation of model sensitivity, a distance power of 0.25 and max rank of 10km was selected for all parameters.

The selected setting yielded the lowest repeated mean absolute error (MAE) for salinity (2.364 ppt). Dissolved oxygen and water temperature had slightly lower MAEs at a distance power of 0.50 and a 10 km range: 0.433 mg/L and 0.955 °C, respectively, compared with 0.443 mg/L and 0.964 °C at the selected settings. Turbidity had its lowest MAE at a distance power of 0.50 and a 5 km range (1.127 NTU), although this combination provided a mean withheld-bin prediction coverage of 99.7%, compared with 100% at the selected setting. Thus, the final settings did not uniformly minimize MAE across the four parameters and overall raster smoothness was favored over absolute lower MAE as such gradients are more ecologically representative of reality.

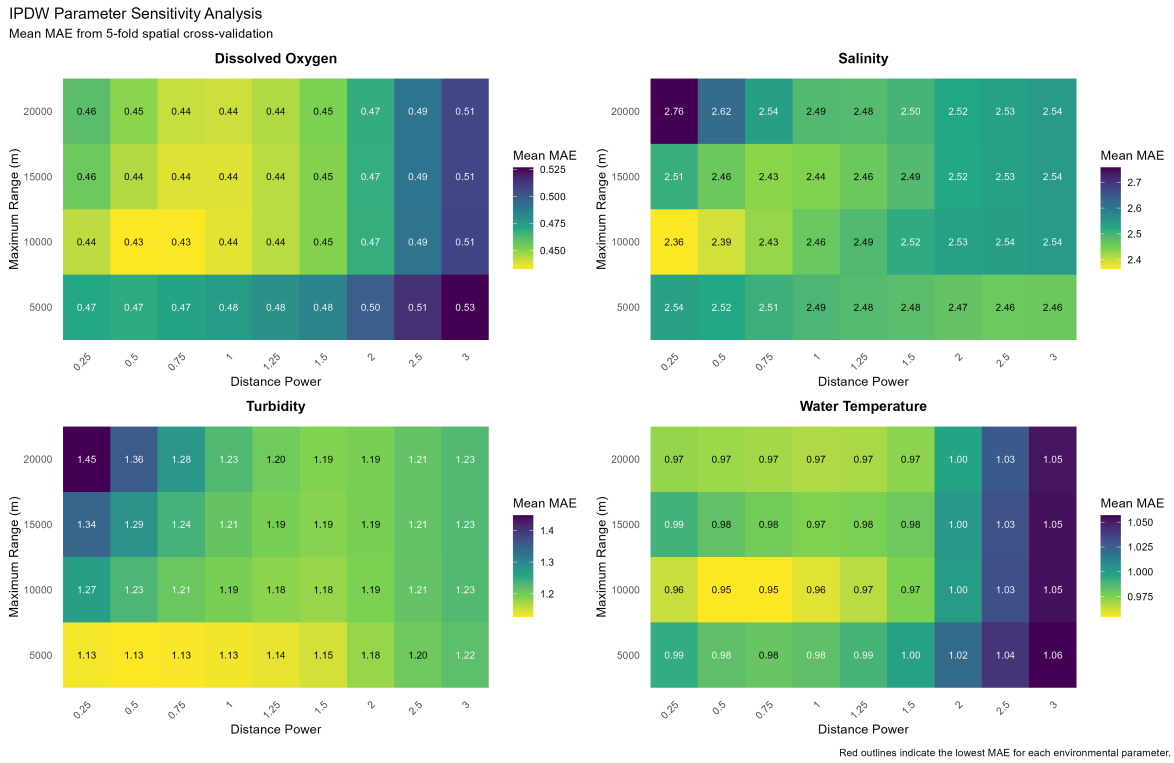


Figure 3: Sensitivity analysis for predictor variables.

Table 5: Repeated spatial cross-validation results for distance power 0.25 and maximum path distance 10 km.

Parameter	Units	Mean MAE (SD)	Mean RMSE (SD)	Mean bias	Mean coverage
Dissolved oxygen	mg/L	0.443 (0.010)	0.565 (0.010)	0.009	100%

Parameter	Units	Mean MAE (SD)	Mean RMSE (SD)	Mean bias	Mean coverage
Salinity	ppt	2.364 (0.039)	3.192 (0.030)	0.197	100%
Turbidity	NTU	1.269 (0.027)	1.684 (0.028)	−0.151	100%
Water temperature	°C	0.964 (0.019)	1.162 (0.051)	−0.067	100%

Environmental Prediction Surface Rasters

The resulting environmental prediction surfaces had a cell size of 100 x 100m. Predicted values were 5.11 to 7.99 mg/L for dissolved oxygen, 2.31 to 26.73 ppt for salinity, 0.03 to 9.62 NTU for turbidity, and 25.93 to 30.42 °C for water temperature.

Dissolved oxygen generally predicted lower to the east and north east, salinity predicted higher to the south west, turbidity was higher in the north west near the mouth of Escambias River, and water temp was lowest near Blackwater Bay and the mouth of Blackwater River.

Discussion

In progress...

Conclusion

In progress...

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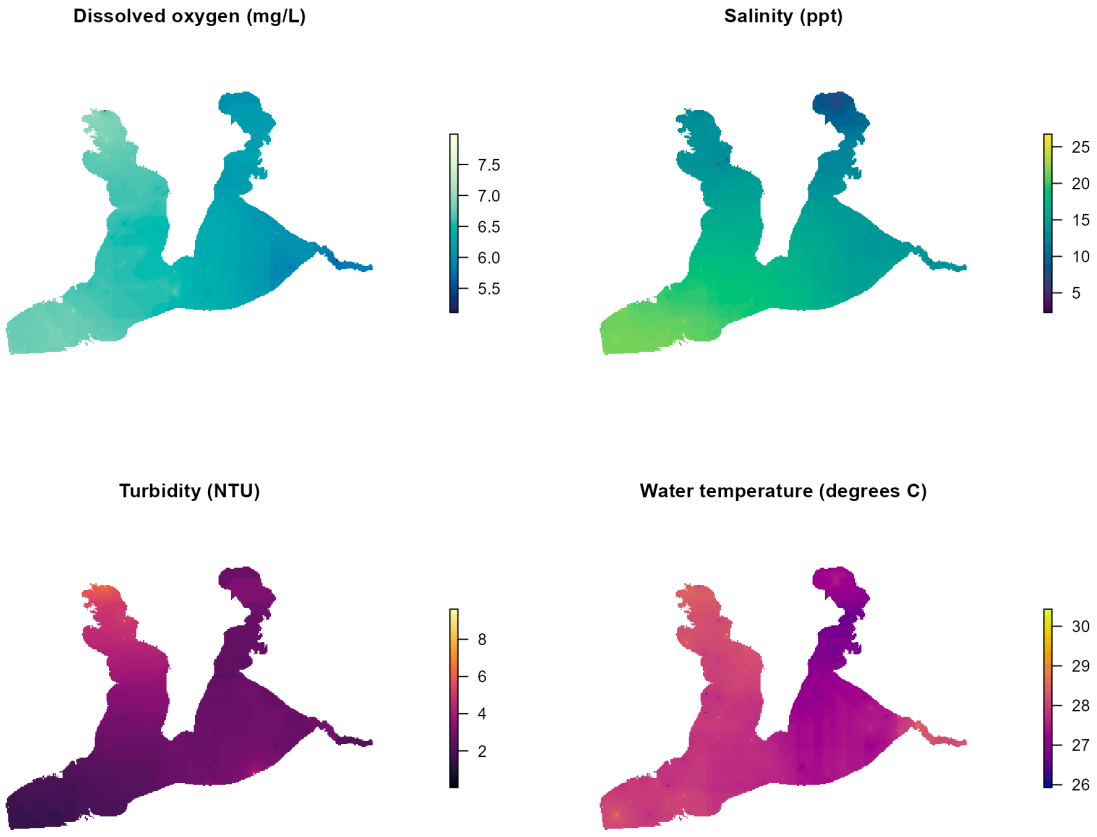


Figure 4: Final prediction rasters

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